

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

MFG302-R

Architect a factory OS for smart manufacturing

Steve Blackwell

WW Tech Lead Manufacturing
AWS



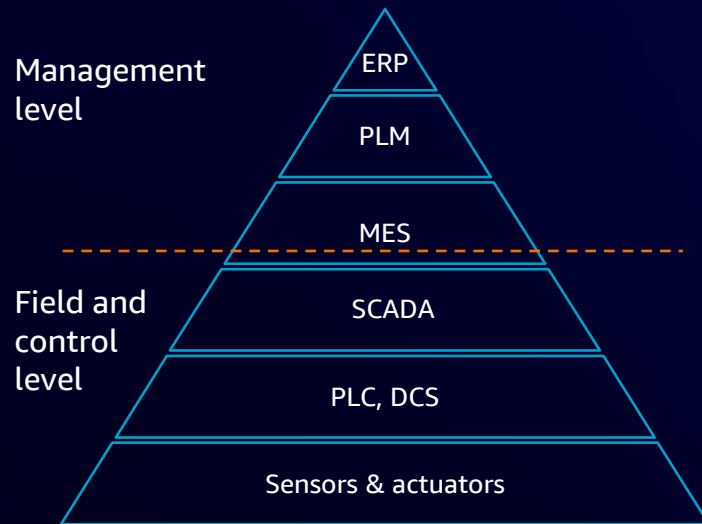
© 2023, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Agenda

- 01 Manufacturing industry transformation
- 02 Smart manufacturing principles
- 03 Solution components
- 04 Whiteboard time

Evolving ISA95 to transform

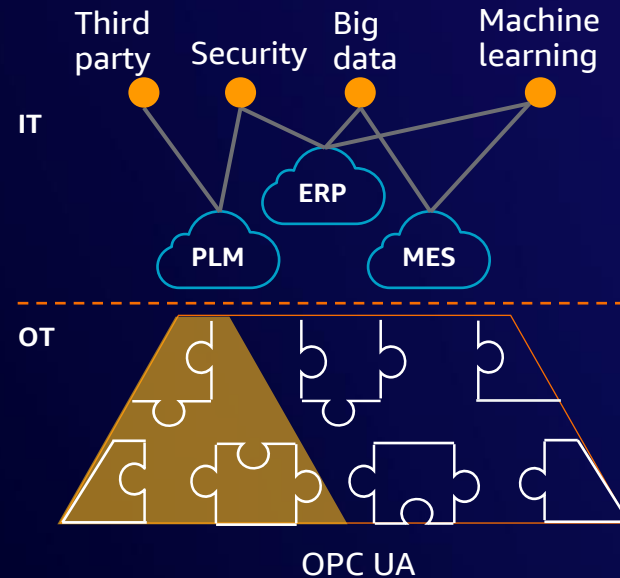
Today



Monolithic Pyramid

- Standalone applications
- Data silos
- Poor upstream/downstream communication
- Disparate proprietary protocols

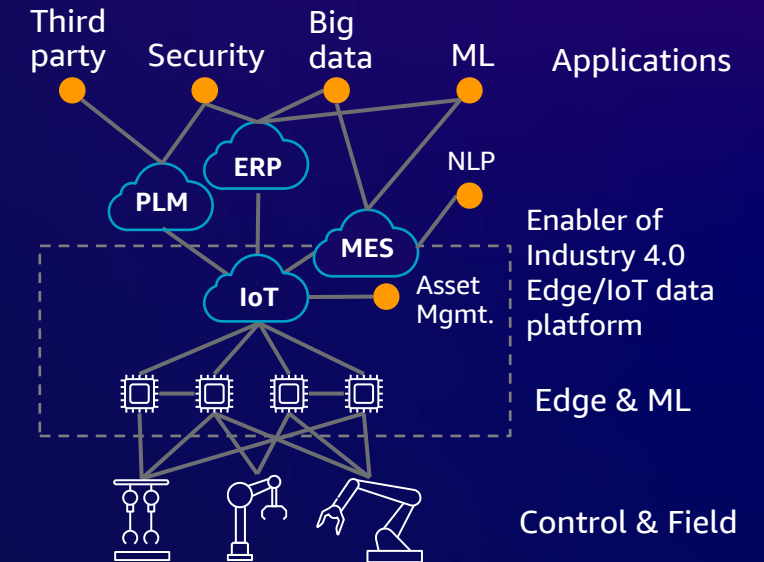
Today/Tomorrow



Converging IT and OT

- Cloud computing revolutionized IT
- Flexible shop floor connectivity
- Descriptive protocols
- System integration

Near Future



Smart Factory

- Border between IT and OT is gone
- Any-to-any communication
- Data transparency
- Edge/Cloud hybrid model

Data-driven solutions

Current systems

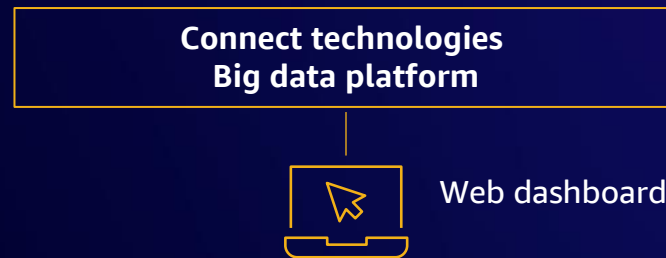
Traditional



Challenges

- Data in silos
- Missing source code
- Limitations in scalability and maintainability
- Serious security weaknesses

Today/Tomorrow



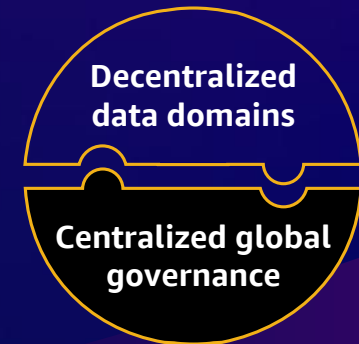
Challenges

- Difficulties in requirement prioritization
- Complex data engineering
- Limited use of managed services



Knowledge-based systems

Open data , API-based



Next gen data-driven companies

Decentralized domain-based

Supply chain, R&D, manufacturing, sustainability, connected products

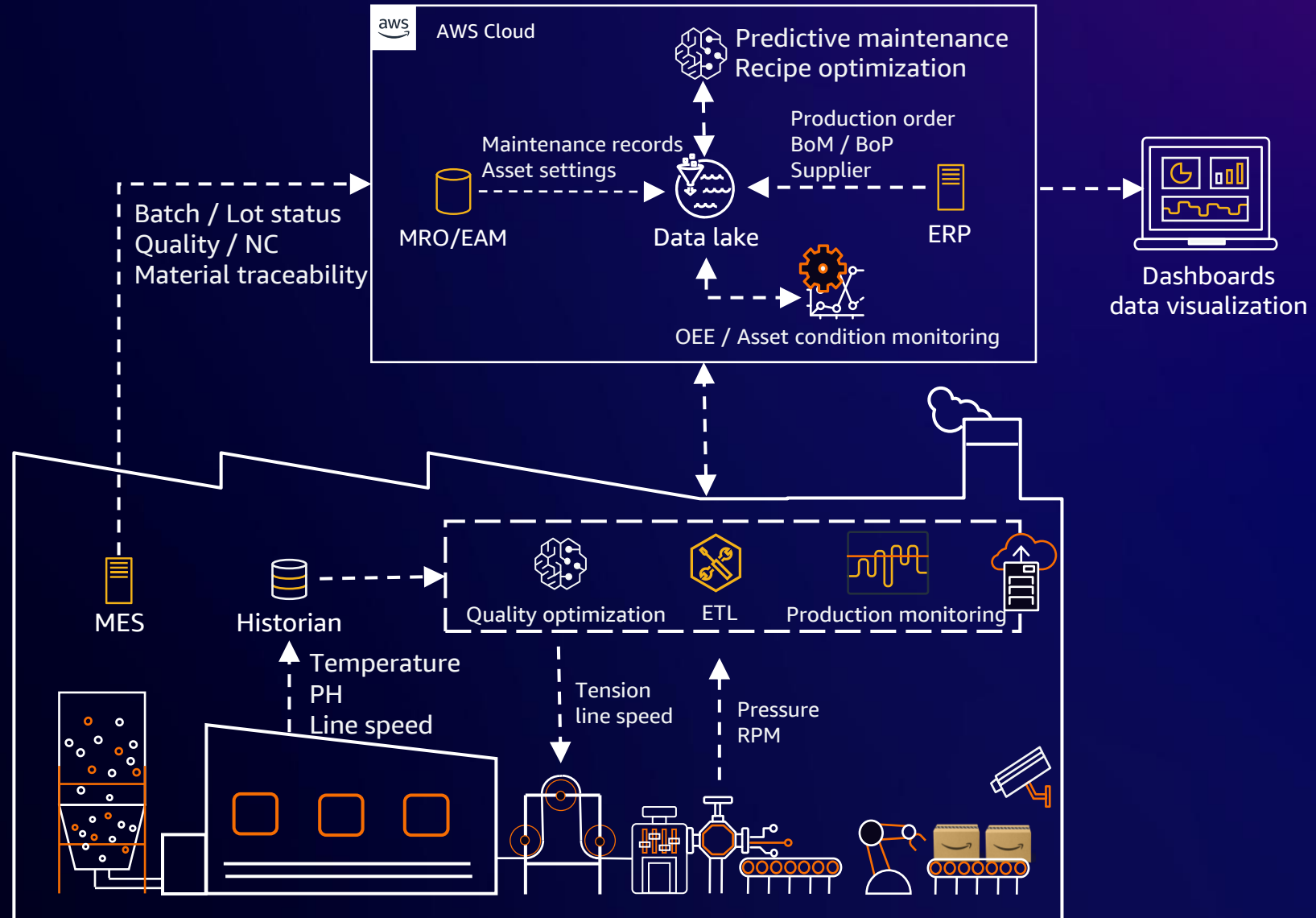
1st gen data-driven companies

Decentralized data silos

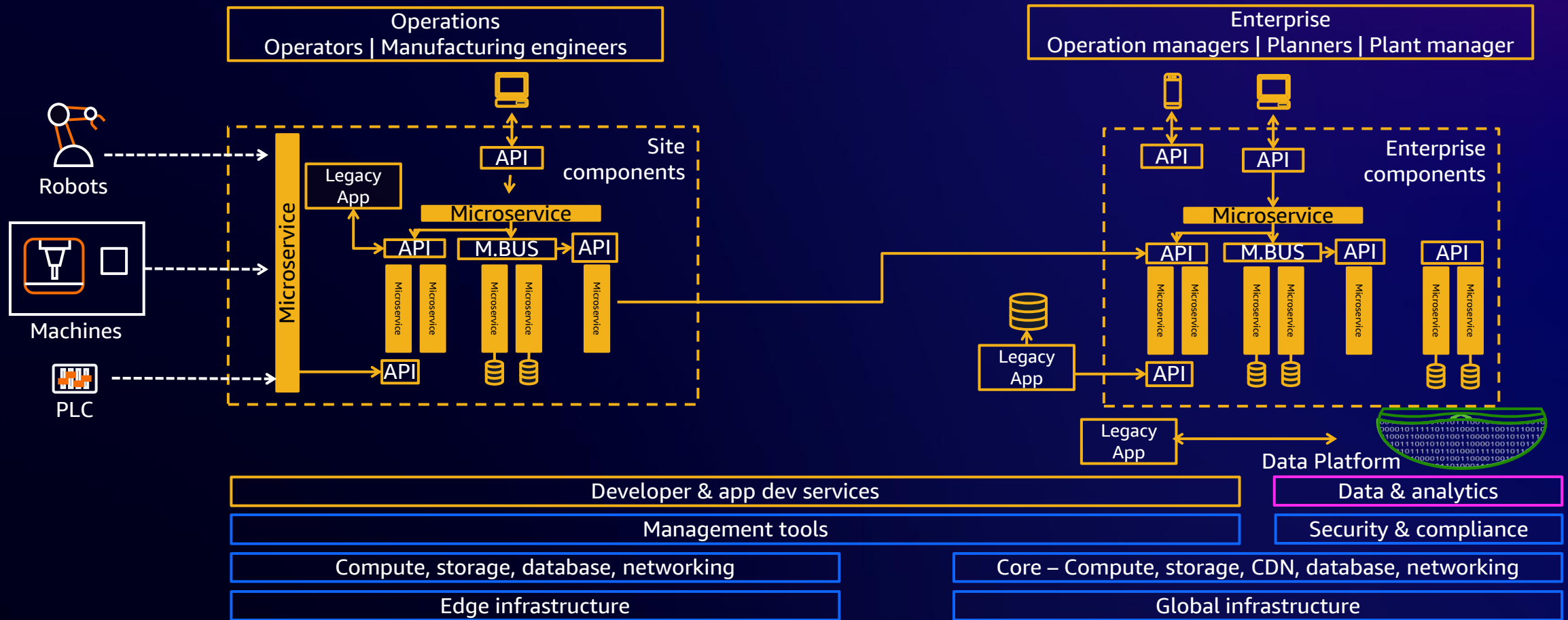
2nd gen data-driven companies

Centralized data platform

Smart manufacturing examples

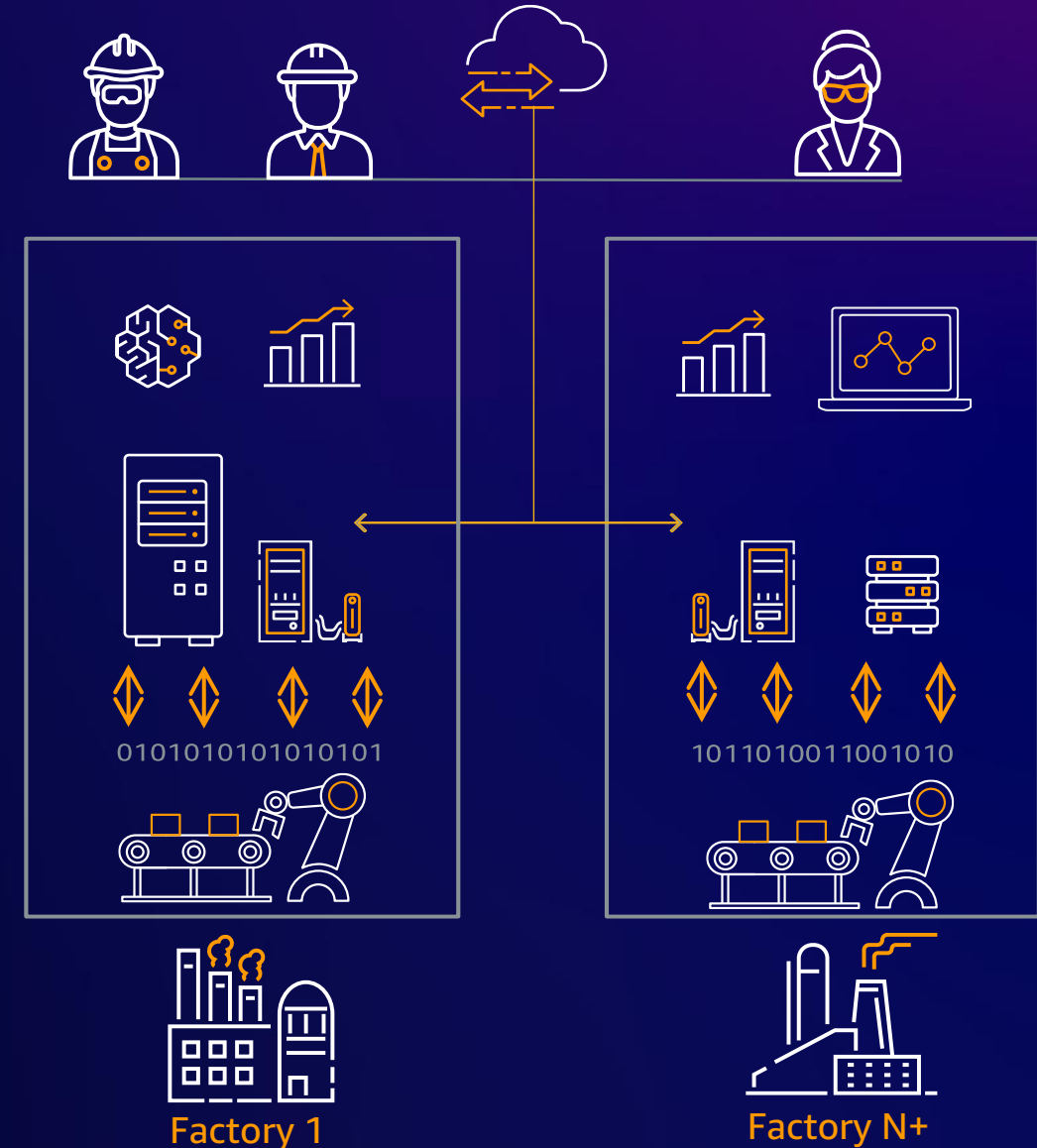


Factory OS Conceptual Architecture



Industrial edge components

- Last meter of connectivity
- Data pipelines
- Single management “pane of glass” across all industrial edges
- Runtime environment for containers and ML models
- Message Bus (NS & EW)
- Secure cloud connection



Industrial edge: Compute and storage



Industrial PC



vSphere
VMWare

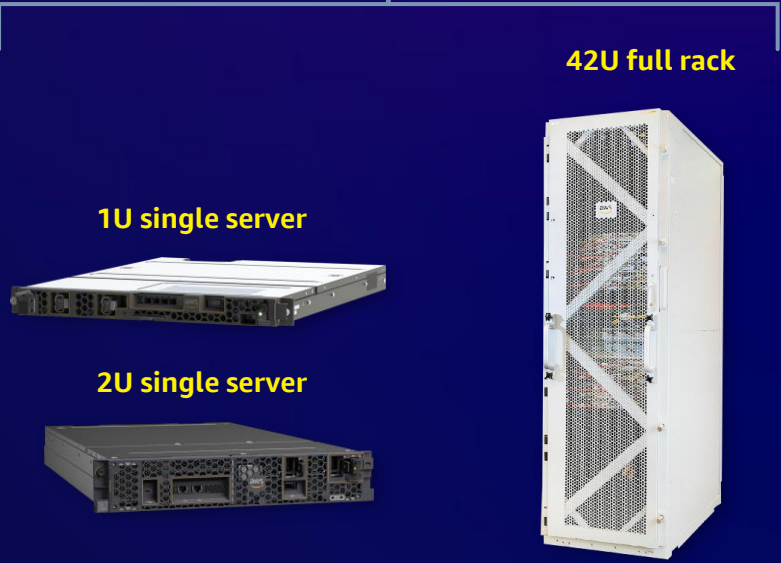


AWS Snowcone



AWS Snowball Edge

AWS Outposts family



1U single server

2U single server

42U full rack

Line side



Data center



Amazon CloudWatch



AWS Systems Manager



AWS IAM



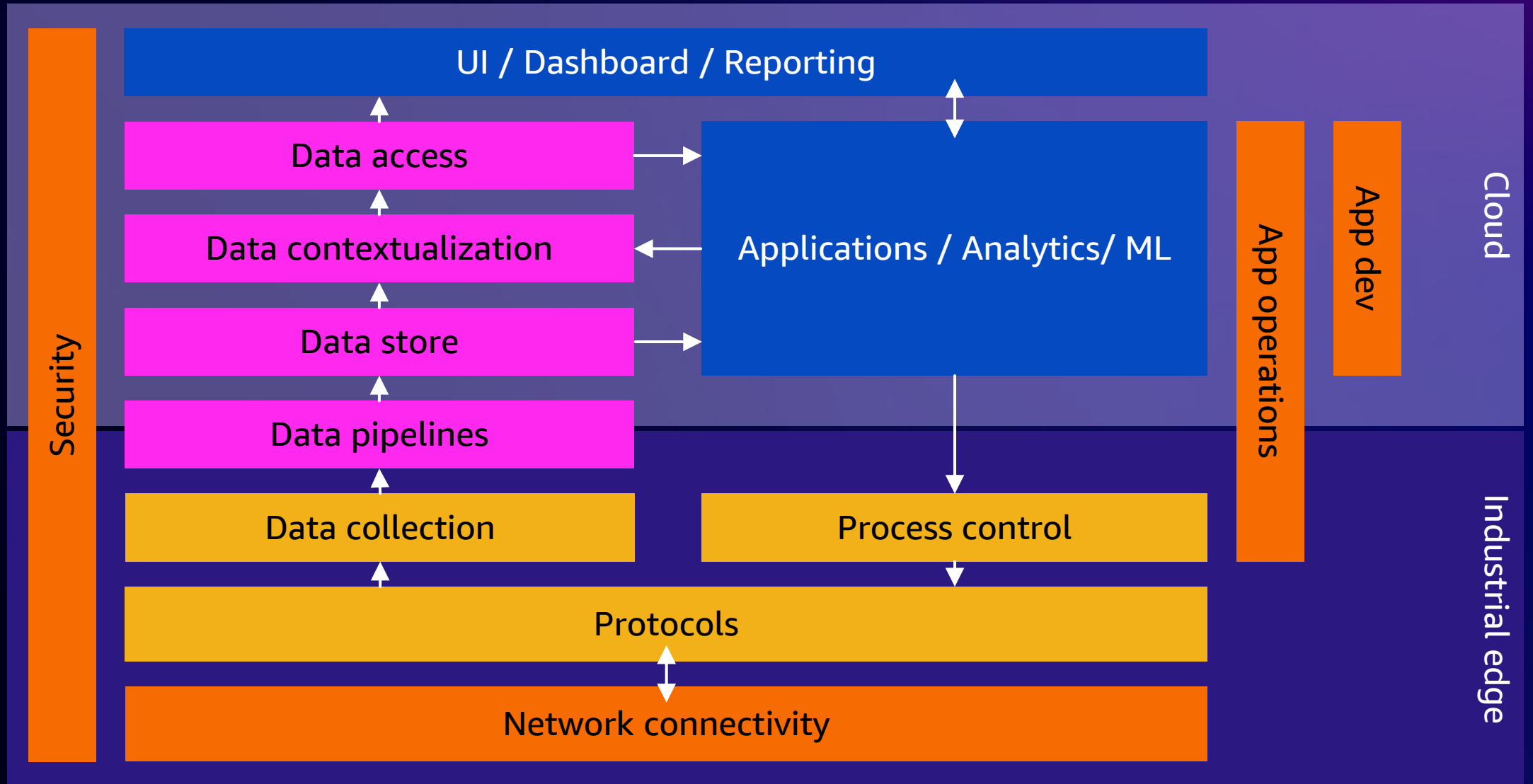
AWS CloudTrail

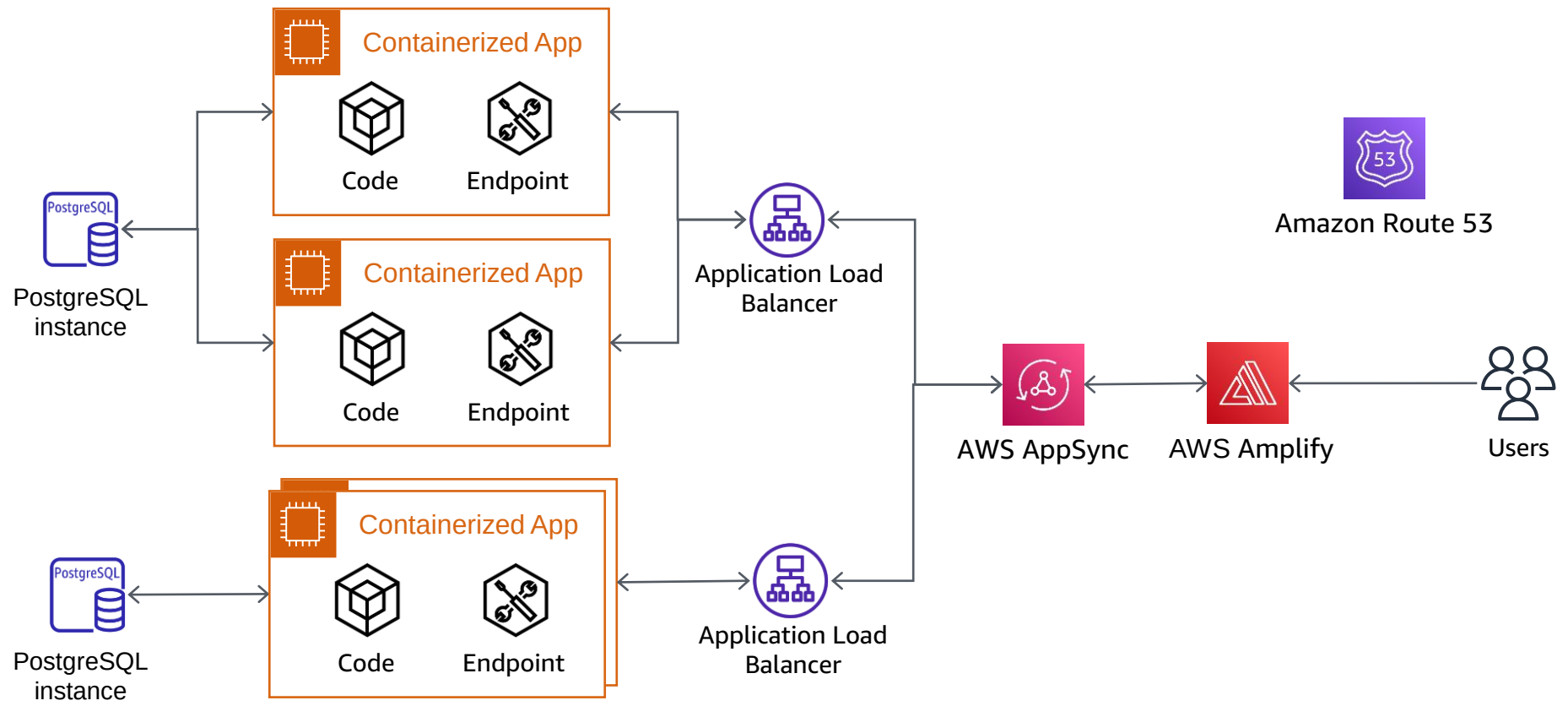
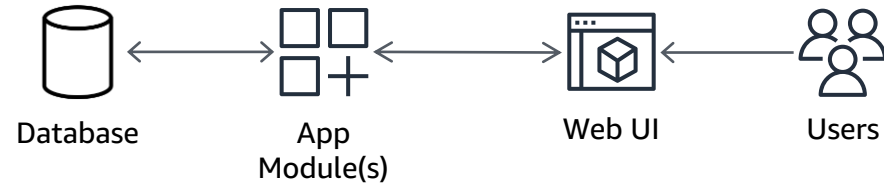


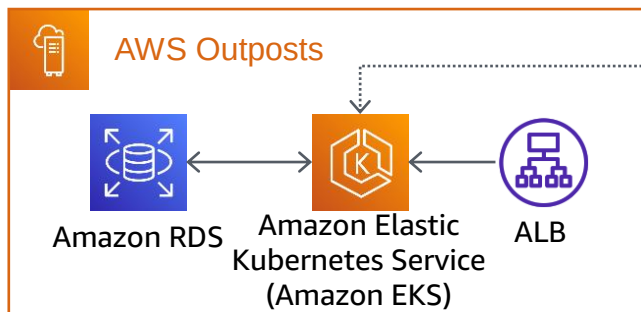
Whiteboarding time



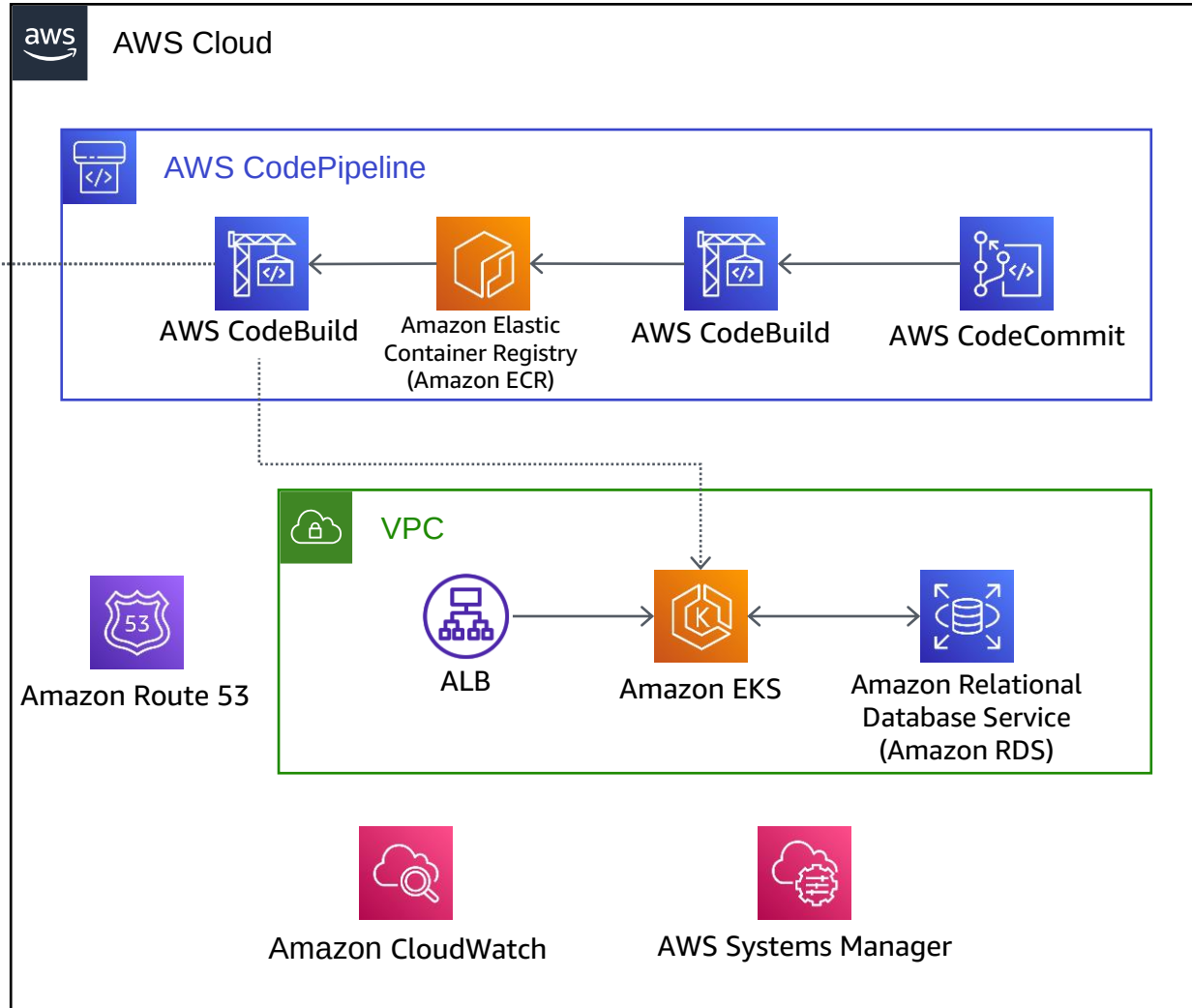
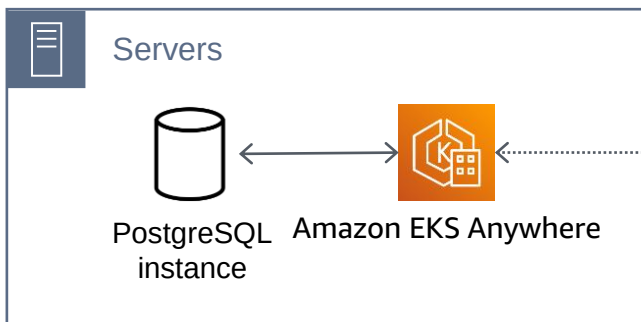
Factory OS architectural blocks



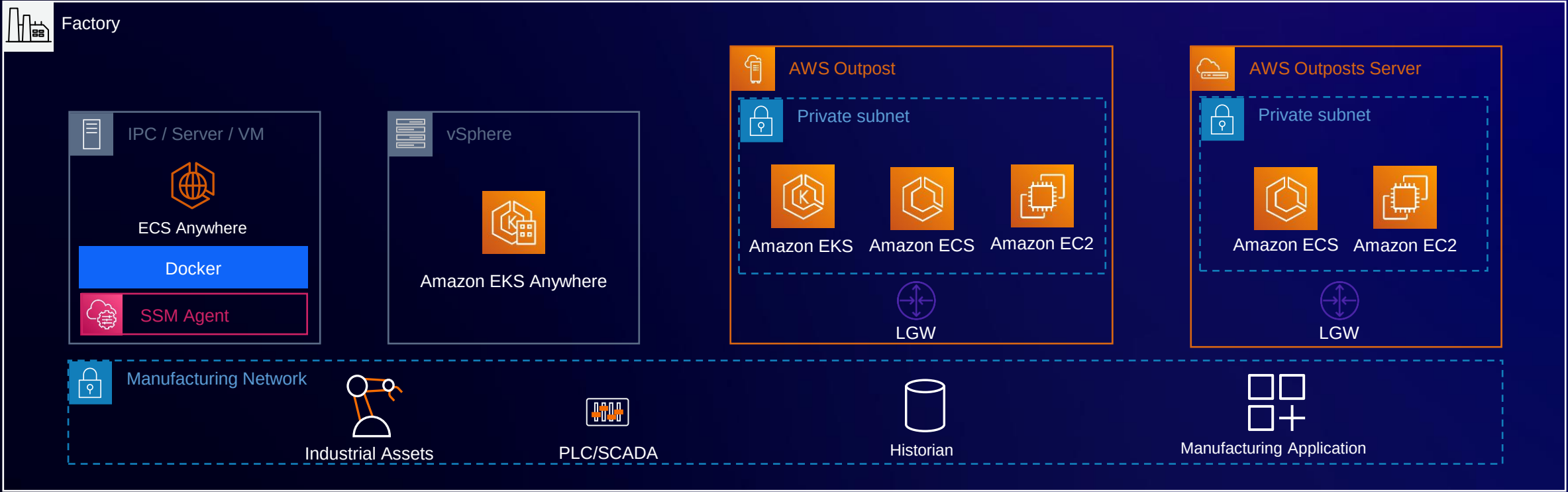




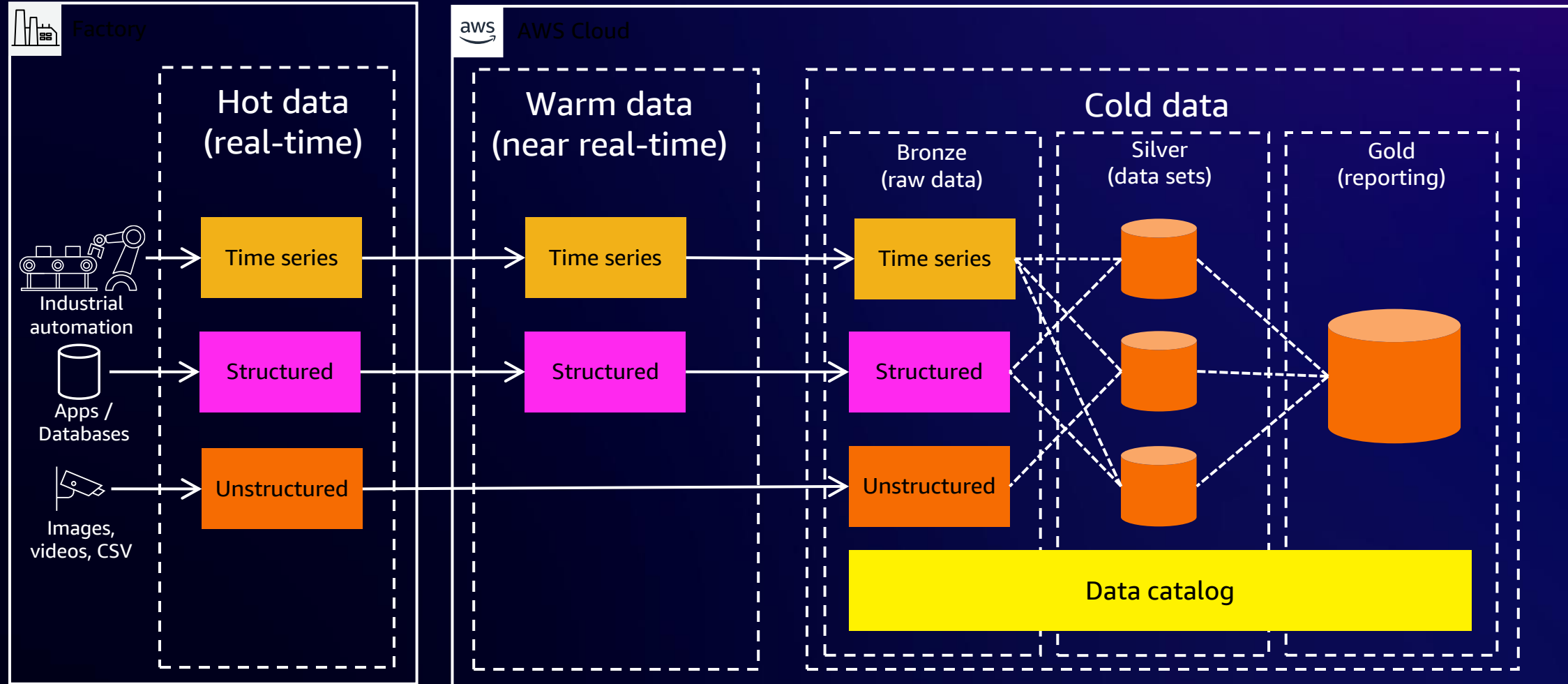
Network Services (DNS)



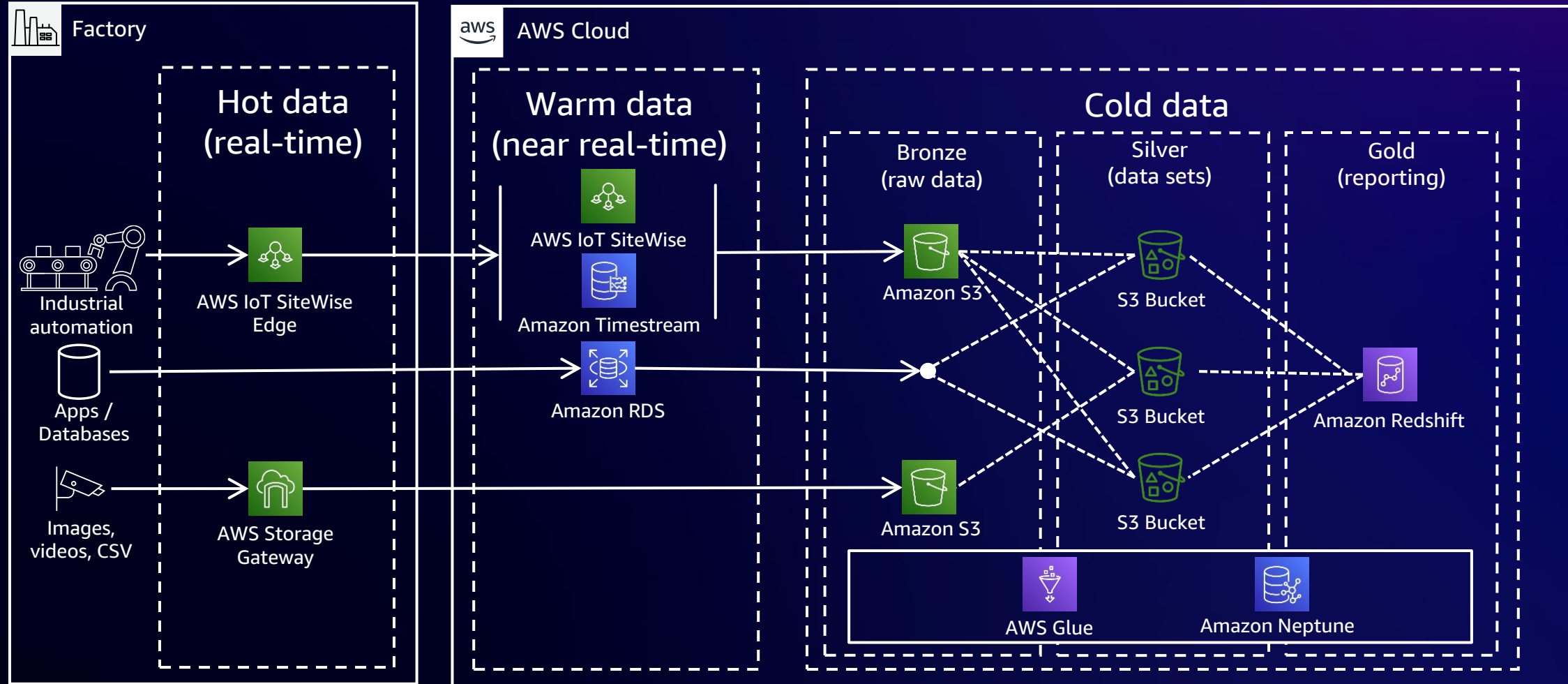
Containers



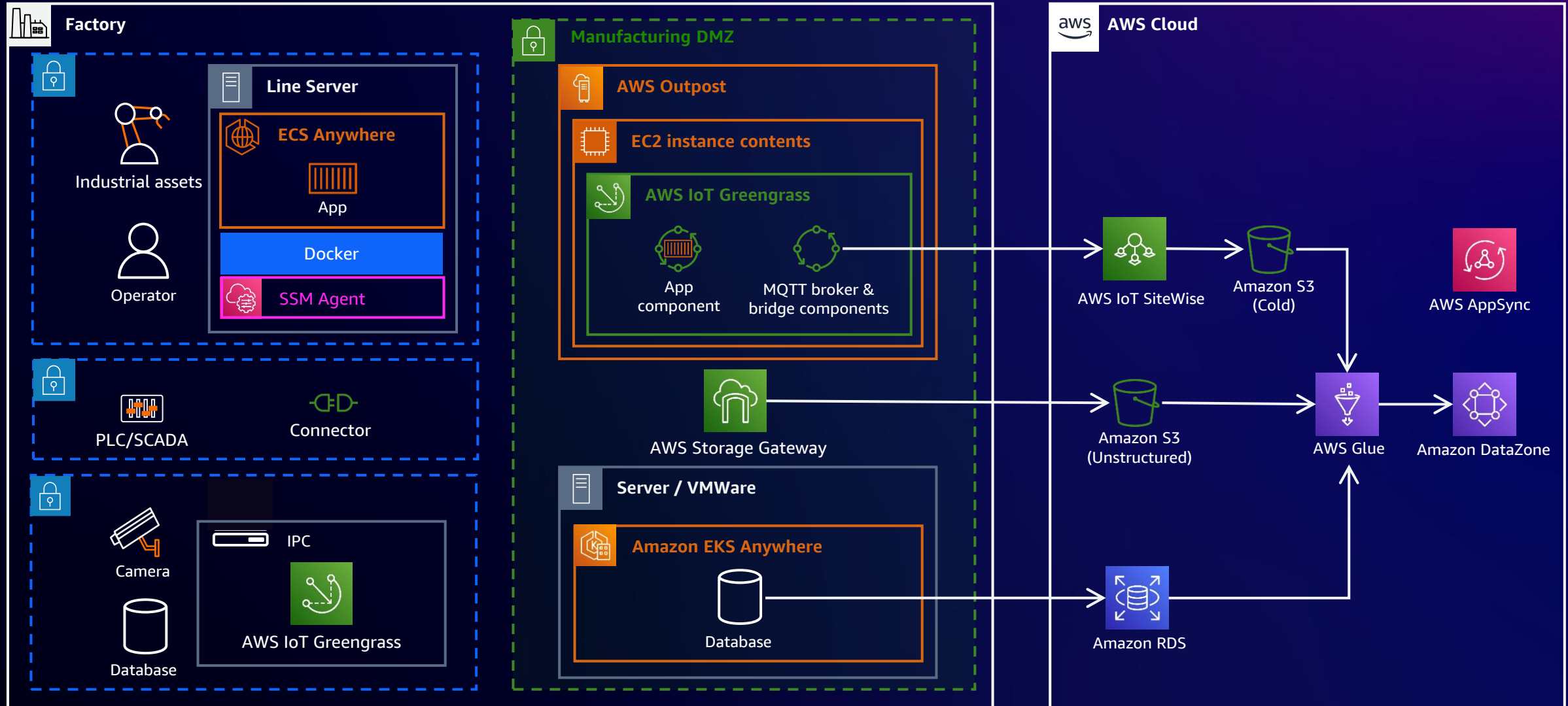
Industrial data architecture



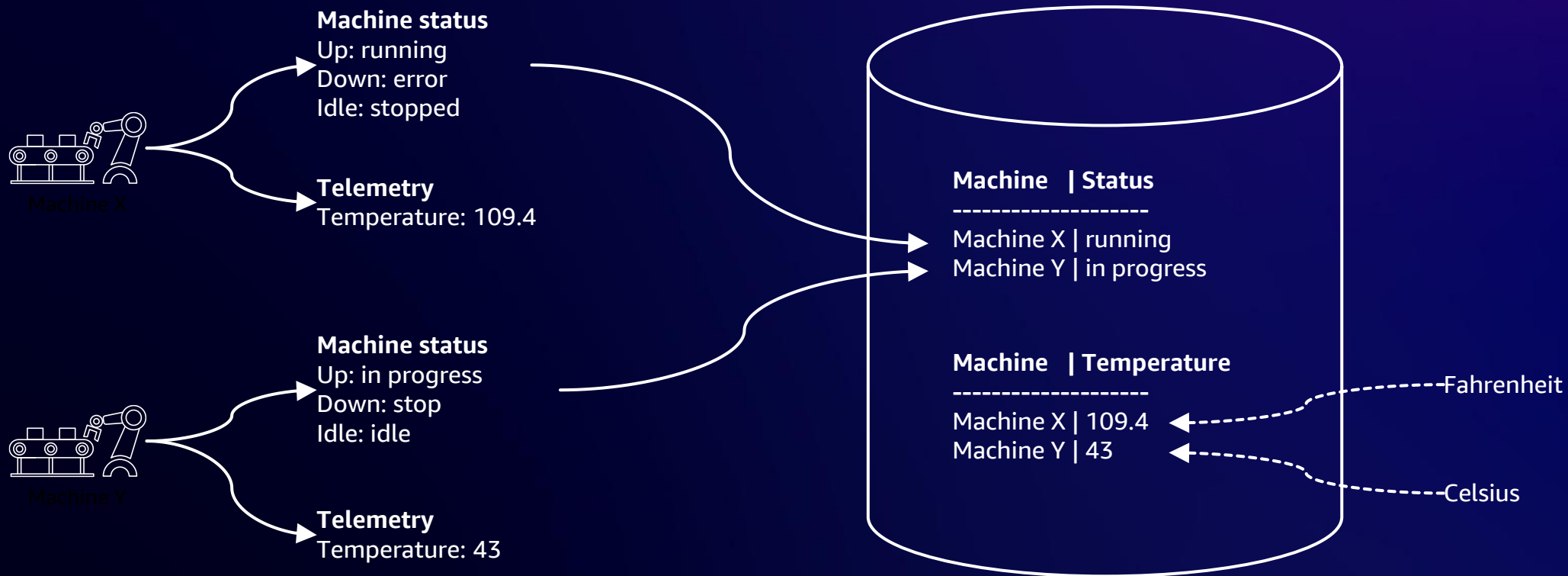
Industrial data **reference** architecture: Example



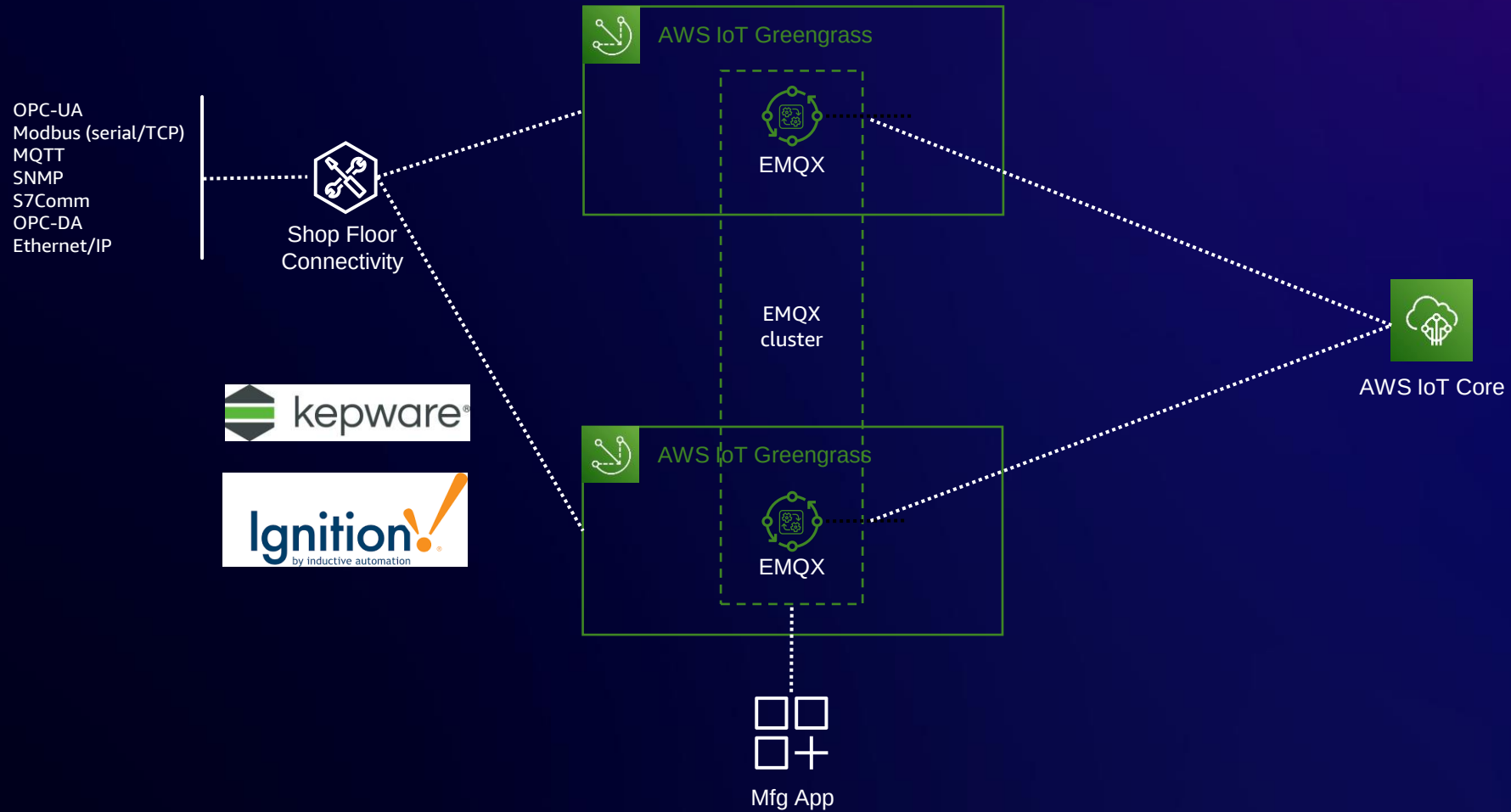
Data ingestion



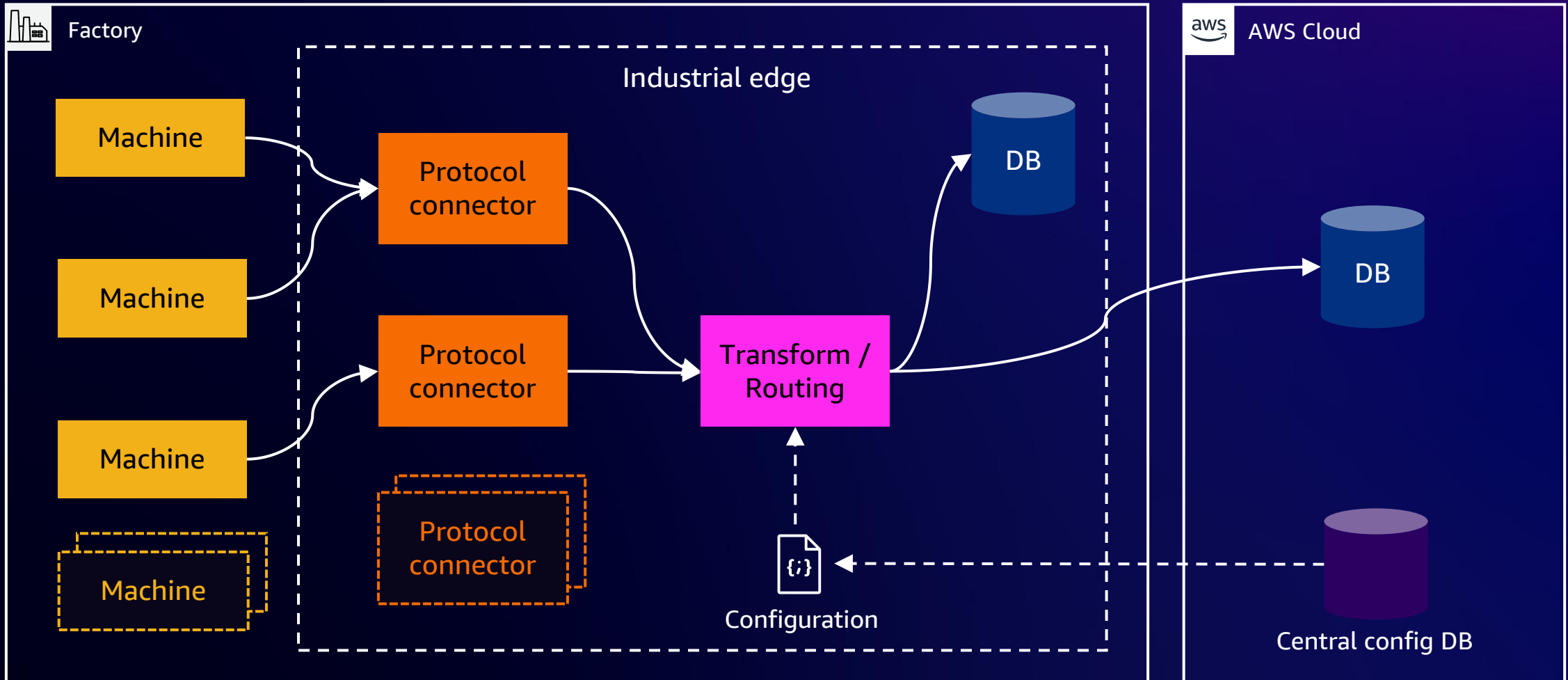
Information model challenges



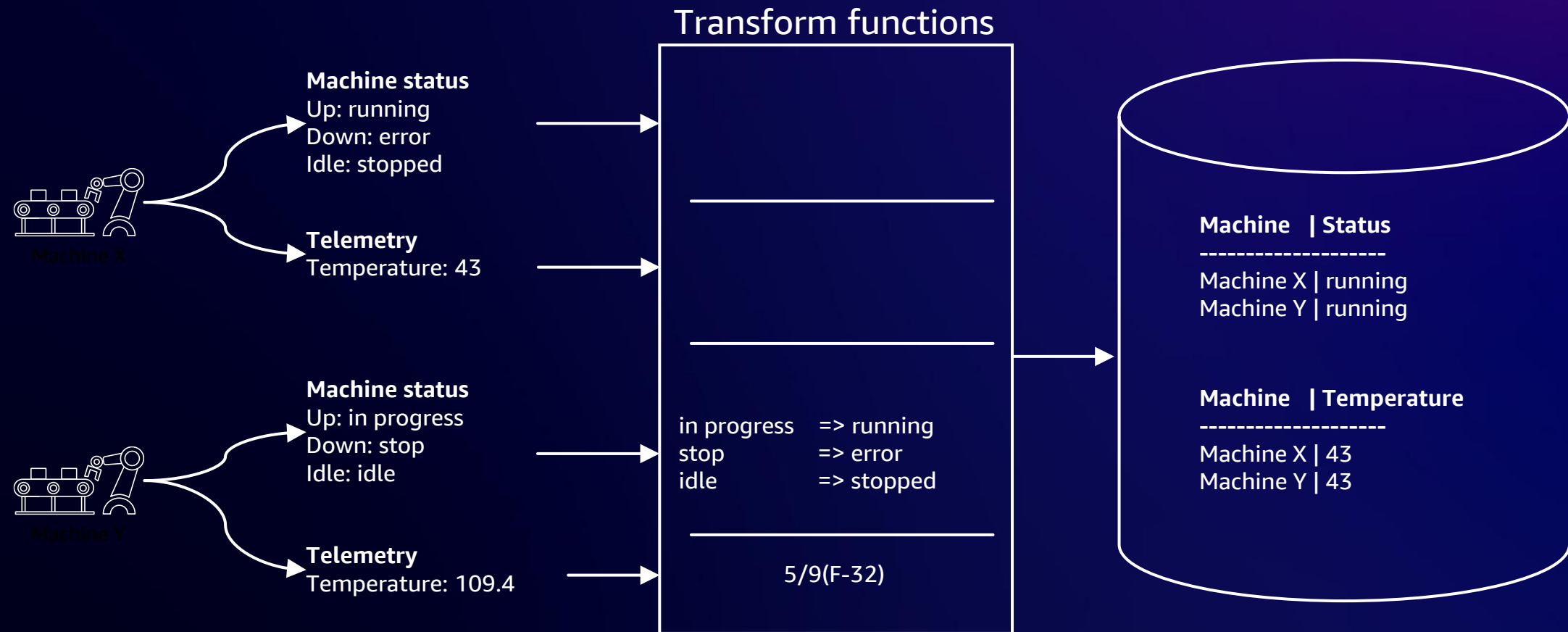
Message Bus



Time series (IIoT) ingestion architecture



Information model example



Thank you!

Steve Blackwell

 [linkedin.com/in/smbblackwell](https://www.linkedin.com/in/smbblackwell)



Please complete the session
survey in the mobile app